Appendix A

Vector differential operators

In this Appendix we introduce orthogonal curvilinear coordinates and derive the general expressions of the vector differential operators in this kind of coordinates. Moreover, we give the expressions of the differential operators for the particular cases of cylindrical and spherical coordinates. The presentation follows closely the treatment given in *Calculus of Several Variables* (Adams [2, p. 336]) from which the figures have been borrowed, by gentle permission of the author.

A.1 Orthogonal curvilinear coordinates

We assume that (u, v, w) are a set of *orthogonal* curvilinear coordinates in xyz-space defined via the transformation

$$x = x(u, v, w), \qquad y = y(u, v, w), \qquad z = z(u, v, w).$$

We also assume that the coordinate surfaces are smooth at any nonsingular point and that the local basis vectors $\widehat{\boldsymbol{u}}$, $\widehat{\boldsymbol{v}}$ and $\widehat{\boldsymbol{w}}$ at any such point form a right-handed triad.

The position vector $\boldsymbol{r}$ of a point P in xyz-space can be expressed in terms of the curvilinear coordinates:

$$\boldsymbol{r} = x(u, v, w)\,\boldsymbol{i} + y(u, v, w)\,\boldsymbol{j} + z(u, v, w)\,\boldsymbol{k}.$$

If we hold $v = v_0$ and $w = w_0$ fixed and let u vary, then $\boldsymbol{r} = \boldsymbol{r}(u, v_0, w_0)$ defines a u-curve in xyz-space. At any point P on this curve, the vector

$$\frac{\partial \boldsymbol{r}}{\partial u} = \frac{\partial x}{\partial u}\,\boldsymbol{i} + \frac{\partial y}{\partial u}\,\boldsymbol{j} + \frac{\partial z}{\partial u}\,\boldsymbol{k}$$

is tangent to the u-curve at P. In general, the three vectors

$$\frac{\partial \boldsymbol{r}}{\partial u}, \qquad \frac{\partial \boldsymbol{r}}{\partial v}, \qquad \frac{\partial \boldsymbol{r}}{\partial w}$$

are tangent, respectively, to the the u-curve, the v-curve and the w-curve through P. They are also normal, respectively, to the u-surface, the v-surface and the w-surface through P, so they are mutually perpendicular. (See Figure A.1) The lengths of these tangent vectors are called the *scale factors* of the coordinate system and are therefore defined by

$$h_u = \left|\frac{\partial \boldsymbol{r}}{\partial u}\right|, \qquad h_v = \left|\frac{\partial \boldsymbol{r}}{\partial v}\right|, \qquad h_w = \left|\frac{\partial \boldsymbol{r}}{\partial w}\right|.$$

The scale factors are nonzero at a nonsingular point P of the coordinate system, so a local basis for the coordinate system at P can be obtained by dividing the tangent vectors to the coordinate curves by their lengths. Denoting the local basis vectors by $\widehat{\boldsymbol{u}}$, $\widehat{\boldsymbol{v}}$ and $\widehat{\boldsymbol{w}}$, we have

$$\frac{\partial \boldsymbol{r}}{\partial u} = h_u \widehat{\boldsymbol{u}}, \qquad \frac{\partial \boldsymbol{r}}{\partial v} = h_v \widehat{\boldsymbol{v}}, \qquad \frac{\partial \boldsymbol{r}}{\partial w} = h_w \widehat{\boldsymbol{w}}.$$

The basis vectors $\widehat{\boldsymbol{u}}$, $\widehat{\boldsymbol{v}}$ and $\widehat{\boldsymbol{w}}$ will form a right-handed triad provided we have chosen a suitable order for the coordinates u, v and w.

The volume element in an orthogonal curvilinear coordinate system is the volume of an infinitesimal *coordinate box* bounded by pairs of u-, v- and w-surfaces corresponding to values u and $u + du$, v and $v + dv$ and w and $w + dw$, respectively. See Figure A.1. Since these coordinate surfaces are assumed smooth, and since they intersect at right angles, the coordinate box is rectangular and is spanned by the vectors

$$\frac{\partial \boldsymbol{r}}{\partial u}\, du = h_u\, du\, \widehat{\boldsymbol{u}}, \qquad \frac{\partial \boldsymbol{r}}{\partial v}\, dv = h_v\, dv\, \widehat{\boldsymbol{v}}, \qquad \frac{\partial \boldsymbol{r}}{\partial w}\, dw = h_w\, dw\, \widehat{\boldsymbol{w}}.$$

Therefore, the volume element is given by

$$dV = h_u h_v h_w\, du\, dv\, dw.$$

Furthermore, the surface area elements on the u-, v- and w-surfaces are the areas of the appropriate faces of the coordinate box:

$$dS_u = h_v h_w\, dv\, dw, \qquad dS_v = h_u h_w\, du\, dw, \qquad dS_w = h_u h_v\, du\, dv.$$

The arc length elements along the u-, v- and w-coordinate curves are the edges of the coordinate box:

$$ds_u = h_u\, du, \qquad ds_v = h_v\, dv, \qquad ds_w = h_w\, dw.$$

A.2 Differential operators

The gradient $\boldsymbol{\nabla} f$ of a scalar field f can be expressed in terms of the local basis at any point P with curvilinear coordinates (u, v, w) in the form

$$\boldsymbol{\nabla} f = F_u\, \widehat{\boldsymbol{u}} + F_v\, \widehat{\boldsymbol{v}} + F_w\, \widehat{\boldsymbol{w}}.$$

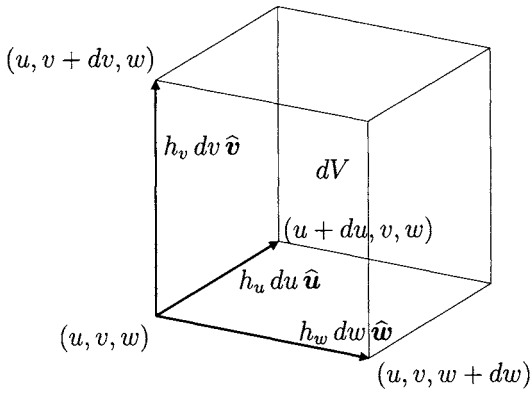

Figure A.1: Infinitesimal coordinate box of an orthogonal coordinate system.

In order to determine the coefficients F_u, F_v and F_w in this formula, we will compare two expressions for the directional derivative of f along an arbitrary curve in xyz-space.

If the curve $\mathcal{C}$ has parametrization $\boldsymbol{r} = \boldsymbol{r}(s)$ in terms of arc length s, then the directional derivative of f along $\mathcal{C}$ is given by

$$\frac{df}{ds} = \frac{\partial f}{\partial u}\frac{du}{ds} + \frac{\partial f}{\partial v}\frac{dv}{ds} + \frac{\partial f}{\partial w}\frac{dw}{ds}.$$

On the other hand, this directional derivative is also given by $\frac{df}{ds} = \boldsymbol{\nabla} f \cdot \hat{\boldsymbol{\gamma}}$, where $\hat{\boldsymbol{\gamma}}$ is the unit tangent vector to $\mathcal{C}$. We have

$$\begin{aligned}\hat{\boldsymbol{\gamma}} = \frac{d\boldsymbol{r}}{ds} &= \frac{\partial \boldsymbol{r}}{\partial u}\frac{du}{ds} + \frac{\partial \boldsymbol{r}}{\partial v}\frac{dv}{ds} + \frac{\partial \boldsymbol{r}}{\partial w}\frac{dw}{ds} \\ &= h_u \frac{du}{ds}\,\hat{\boldsymbol{u}} + h_v \frac{dv}{ds}\,\hat{\boldsymbol{v}} + h_w \frac{dw}{ds}\,\hat{\boldsymbol{w}}.\end{aligned}$$

Thus

$$\frac{df}{ds} = \boldsymbol{\nabla} f \cdot \hat{\boldsymbol{\gamma}} = F_u h_u \frac{du}{ds} + F_v h_v \frac{dv}{ds} + F_w h_w \frac{dw}{ds}.$$

Comparing these two expressions for df/ds along $\mathcal{C}$, we see that

$$F_u h_u = \frac{\partial f}{\partial u}, \qquad F_v h_v = \frac{\partial f}{\partial v}, \qquad F_w h_w = \frac{\partial f}{\partial w}.$$

Therefore, we have shown that

$$\boldsymbol{\nabla} f(u, v, w) = \frac{1}{h_u}\frac{\partial f}{\partial u}\,\hat{\boldsymbol{u}} + \frac{1}{h_v}\frac{\partial f}{\partial v}\,\hat{\boldsymbol{v}} + \frac{1}{h_w}\frac{\partial f}{\partial w}\,\hat{\boldsymbol{w}}.$$

Now consider a vector field $\boldsymbol{F}$ expressed in terms of the curvilinear coordinates:

$$\boldsymbol{F}(u,v,w) = F_u(u,v,w)\,\hat{\boldsymbol{u}} + F_v(u,v,w)\,\hat{\boldsymbol{v}} + F_w(u,v,w)\,\hat{\boldsymbol{w}}.$$

The flux of $\boldsymbol{F}$ out of the infinitesimal coordinate box of Figure A.1 is the sum of the fluxes of $\boldsymbol{F}$ out of the three pairs of opposite surfaces of the box. The flux out of the u-surfaces corresponding to u and $u+du$ is

$$\begin{aligned}
&\boldsymbol{F}(u+du,v,w)\cdot\hat{\boldsymbol{u}}\,dS_u - \boldsymbol{F}(u,v,w)\cdot\hat{\boldsymbol{u}}\,dS_u \\
&\qquad = (F_u(u+du,v,w)\,h_v(u+du,v,w)\,h_w(u+du,v,w) \\
&\qquad\qquad -F_u(u,v,w)\,h_v(u,v,w)\,h_w(u,v,w))\,dv\,dw \\
&\qquad = \frac{\partial}{\partial u}(h_v h_w F_u)\,du\,dv\,dw.
\end{aligned}$$

Similar expressions hold for the fluxes out of the other pairs of coordinate surfaces.

The divergence at P of $\boldsymbol{F}$ is the flux *per unit volume* out of the infinitesimal coordinate box at P. Thus it is given by

$$\begin{aligned}
\boldsymbol{\nabla}\cdot\boldsymbol{F}(u,v,w) = \frac{1}{h_u h_v h_w}\Big[&\frac{\partial}{\partial u}(h_v h_w F_u(u,v,w)) + \frac{\partial}{\partial v}(h_u h_w F_v(u,v,w)) \\
&+\frac{\partial}{\partial w}(h_u h_v F_w(u,v,w))\Big].
\end{aligned}$$

To calculate the curl of a vector field expressed in terms of orthogonal curvilinear coordinates we can make use of some previously obtained vector identities. First, observe that the gradient of the scalar field $f(u,v,w) = u$ is $\hat{\boldsymbol{u}}/h_u$, so that $\hat{\boldsymbol{u}} = h_u\boldsymbol{\nabla}u$. Similarly, $\hat{\boldsymbol{v}} = h_v\boldsymbol{\nabla}v$ and $\hat{\boldsymbol{w}} = h_w\boldsymbol{\nabla}w$. Therefore, the vector field

$$\boldsymbol{F} = F_u\,\hat{\boldsymbol{u}} + F_v\,\hat{\boldsymbol{v}} + F_w\,\hat{\boldsymbol{w}}$$

can be written in the form

$$\boldsymbol{F} = F_u h_u\boldsymbol{\nabla}u + F_v h_v\boldsymbol{\nabla}v + F_w h_w\boldsymbol{\nabla}w.$$

Using the identity $\boldsymbol{\nabla}\times(f\boldsymbol{\nabla}g) = \boldsymbol{\nabla}f\times\boldsymbol{\nabla}g$, we can calculate the curl of each term in the expression above. We have

$$\begin{aligned}
\boldsymbol{\nabla}\times(F_u h_u\boldsymbol{\nabla}u) &= \boldsymbol{\nabla}(F_u h_u)\times\boldsymbol{\nabla}u \\
&= \left[\frac{1}{h_u}\frac{\partial}{\partial u}(F_u h_u)\,\hat{\boldsymbol{u}} + \frac{1}{h_v}\frac{\partial}{\partial v}(F_u h_u)\,\hat{\boldsymbol{v}} + \frac{1}{h_w}\frac{\partial}{\partial w}(F_u h_u)\,\hat{\boldsymbol{w}}\right]\times\frac{\hat{\boldsymbol{u}}}{h_u} \\
&= \frac{1}{h_u h_w}\frac{\partial}{\partial w}(F_u h_u)\,\hat{\boldsymbol{v}} - \frac{1}{h_u h_v}\frac{\partial}{\partial v}(F_u h_u)\,\hat{\boldsymbol{w}} \\
&= \frac{1}{h_u h_v h_w}\left[\frac{\partial}{\partial w}(F_u h_u)\,(h_v\hat{\boldsymbol{v}}) - \frac{\partial}{\partial v}(F_u h_u)\,(h_w\hat{\boldsymbol{w}})\right].
\end{aligned}$$

We have used the facts that $\hat{\boldsymbol{u}}\times\hat{\boldsymbol{u}} = \boldsymbol{0}$, $\hat{\boldsymbol{v}}\times\hat{\boldsymbol{u}} = -\hat{\boldsymbol{w}}$ and $\hat{\boldsymbol{w}}\times\hat{\boldsymbol{u}} = \hat{\boldsymbol{v}}$ to obtain the result above. This is why we assumed that the curvilinear coordinate system was right-handed. Corresponding expressions can be calculated for the other two terms in the formula for $\boldsymbol{\nabla}\times\boldsymbol{F}$.

Combining the three terms, we conclude that the curl of

$$\boldsymbol{F} = F_u\,\hat{\boldsymbol{u}} + F_v\,\hat{\boldsymbol{v}} + F_w\,\hat{\boldsymbol{w}}$$

is given by

$$\boldsymbol{\nabla}\times\boldsymbol{F}(u,v,w) = \frac{1}{h_u h_v h_w}\begin{vmatrix} h_u\hat{\boldsymbol{u}} & h_v\hat{\boldsymbol{v}} & h_w\hat{\boldsymbol{w}} \\ \dfrac{\partial}{\partial u} & \dfrac{\partial}{\partial v} & \dfrac{\partial}{\partial w} \\ F_u h_u & F_v h_v & F_w h_w \end{vmatrix}.$$

A.3 Cylindrical coordinates

A.3.1 Definition

For cylindrical coordinates we have $\boldsymbol{r} = r\cos\phi\,\boldsymbol{i} + r\sin\phi\,\boldsymbol{j} + z\,\boldsymbol{k}$, so

$$\frac{\partial\boldsymbol{r}}{\partial r} = \cos\phi\,\boldsymbol{i} + \sin\phi\,\boldsymbol{j}, \qquad \frac{\partial\boldsymbol{r}}{\partial\phi} = -r\sin\phi\,\boldsymbol{i} + r\cos\phi\,\boldsymbol{j}, \qquad \frac{\partial\boldsymbol{r}}{\partial z} = \boldsymbol{k}.$$

It should be noted that here r does not denote the magnitude of the position vector $\boldsymbol{r}$, but the distance of $\boldsymbol{r}$ from the z-axis. Similarly, the vector $\hat{\boldsymbol{r}}(\phi)$ does not represent the unit vector in direction of $\boldsymbol{r}$, but the directions orthogonal to the z-axis.

The scale factors for the cylindrical coordinate system are given by

$$h_r = \left|\frac{\partial\boldsymbol{r}}{\partial r}\right| = 1, \qquad h_\phi = \left|\frac{\partial\boldsymbol{r}}{\partial\phi}\right| = r, \qquad h_z = \left|\frac{\partial\boldsymbol{r}}{\partial z}\right| = 1,$$

and the local basis consists of the vectors

$$\hat{\boldsymbol{r}} = \cos\phi\,\boldsymbol{i} + \sin\phi\,\boldsymbol{j}, \qquad \hat{\boldsymbol{\phi}} = -\sin\phi\,\boldsymbol{i} + \cos\phi\,\boldsymbol{j}, \qquad \hat{\boldsymbol{z}} = \boldsymbol{k}.$$

See Figure A.2. The local basis is right-handed. The unit vectors do not change with r or z, but

$$\frac{\partial\hat{\boldsymbol{r}}}{\partial\phi} = \hat{\boldsymbol{\phi}}, \qquad \frac{\partial\hat{\boldsymbol{\phi}}}{\partial\phi} = -\hat{\boldsymbol{r}}, \qquad \frac{\partial\hat{\boldsymbol{z}}}{\partial\phi} = 0.$$

The volume element is given by

$$dV = h_r h_\phi h_z\,dr\,d\phi\,dz = r\,dr\,d\phi\,dz.$$

Its boundary consists of surface elements on the cylinder r = constant, the half-plane ϕ = constant and the plane z = constant. These elements are given, respectively, by

$$dS_r = r\,d\phi\,dz, \qquad dS_\phi = dr\,dz, \qquad dS_z = r\,dr\,d\phi.$$

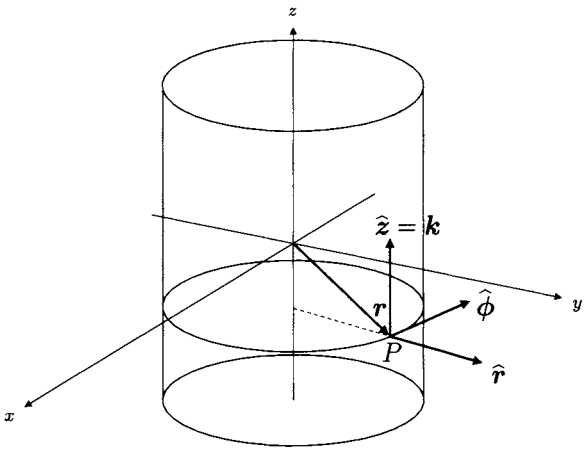

Figure A.2: Cylindrical coordinate system.

A.3.2 Gradient, divergence and curl

In terms of cylindrical coordinates, the gradient of the scalar field $f(r, \phi, z)$ is given by

$$\nabla f(r, \phi, z) = \frac{\partial f}{\partial r}\,\hat{r} + \frac{1}{r}\frac{\partial f}{\partial \phi}\,\hat{\phi} + \frac{\partial f}{\partial z}\,\hat{z}.$$

Since for cylindrical coordinates $h_r = h_z = 1$ and $h_\theta = r$, the divergence of $\boldsymbol{F} = F_r\,\hat{r} + F_\phi\,\hat{\phi} + F_z\,\hat{z}$ is

$$\nabla \cdot \boldsymbol{F} = \frac{1}{r}\left[\frac{\partial}{\partial r}(rF_r) + \frac{\partial}{\partial \phi}F_\phi + \frac{\partial}{\partial z}(rF_z)\right]$$

$$= \frac{\partial F_r}{\partial r} + \frac{1}{r}F_r + \frac{1}{r}\frac{\partial F_\phi}{\partial \phi} + \frac{\partial F_z}{\partial z}.$$

The curl of $\boldsymbol{F}$ is given by

$$\nabla \times \boldsymbol{F} = \frac{1}{r}\begin{vmatrix} \hat{r} & r\,\hat{\phi} & \hat{z} \\ \dfrac{\partial}{\partial r} & \dfrac{\partial}{\partial \phi} & \dfrac{\partial}{\partial z} \\ F_r & rF_\phi & F_z \end{vmatrix}$$

$$= \left(\frac{1}{r}\frac{\partial F_z}{\partial \phi} - \frac{\partial F_\phi}{\partial z}\right)\hat{r} + \left(\frac{\partial F_r}{\partial z} - \frac{\partial F_z}{\partial r}\right)\hat{\phi} + \left(\frac{\partial F_\phi}{\partial r} + \frac{F_\phi}{r} - \frac{1}{r}\frac{\partial F_r}{\partial \phi}\right)\hat{z}.$$

A.3.3 Laplace and advection operators

Remembering that the Laplacian of a function f is defined by the relationship

$$\nabla^2 f = \boldsymbol{\nabla} \cdot \boldsymbol{\nabla} f,$$

the results just obtained for the gradient and the divergence lead to the following expression for the Laplacian in cylindrical coordinates

$$\begin{aligned}\nabla^2 f &= \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial f}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 f}{\partial \phi^2} + \frac{\partial^2 f}{\partial z^2} \\ &= \frac{\partial^2 f}{\partial r^2} + \frac{1}{r}\frac{\partial f}{\partial r} + \frac{1}{r^2}\frac{\partial^2 f}{\partial \phi^2} + \frac{\partial^2 f}{\partial z^2}.\end{aligned}$$

As far as the Laplacian of a vector field $\boldsymbol{F}$ is concerned, its explicit expression can be derived from the vector identity

$$\nabla^2 \boldsymbol{F} = -\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{F}) + \boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \boldsymbol{F}),$$

and using some previously obtained results. We obtain

$$\begin{aligned}\nabla^2 \boldsymbol{F} = &\left(\nabla^2 F_r - \frac{F_r}{r^2} - \frac{2}{r^2}\frac{\partial F_\phi}{\partial \phi}\right)\widehat{\boldsymbol{r}} \\ &+ \left(\nabla^2 F_\phi - \frac{F_\phi}{r^2} + \frac{2}{r^2}\frac{\partial F_r}{\partial \phi}\right)\widehat{\boldsymbol{\phi}} \\ &+ \left(\nabla^2 F_z\right)\widehat{\boldsymbol{z}}.\end{aligned}$$

Finally, we give the expression in cylindrical coordinates of the advection operators for both a scalar and a vector field:

$$\boldsymbol{a} \cdot \boldsymbol{\nabla} u = a_r \frac{\partial u}{\partial r} + \frac{a_\phi}{r}\frac{\partial u}{\partial \phi} + a_z \frac{\partial u}{\partial z},$$

$$\begin{aligned}(\boldsymbol{a} \cdot \boldsymbol{\nabla})\boldsymbol{u} = &\left[(\boldsymbol{a} \cdot \boldsymbol{\nabla})u_r - \frac{a_\phi u_\phi}{r}\right]\widehat{\boldsymbol{r}} \\ &+ \left[(\boldsymbol{a} \cdot \boldsymbol{\nabla})u_\phi + \frac{a_\phi u_r}{r}\right]\widehat{\boldsymbol{\phi}} \\ &+ \left[(\boldsymbol{a} \cdot \boldsymbol{\nabla})u_z\right]\widehat{\boldsymbol{z}}.\end{aligned}$$

A.4 Spherical coordinates

A.4.1 Definition

For spherical coordinates we have

$$\boldsymbol{r} = r\sin\theta\cos\phi\,\boldsymbol{i} + r\sin\theta\sin\phi\,\boldsymbol{j} + r\cos\theta\,\widehat{\boldsymbol{z}}.$$

Thus the tangent vectors to the coordinate curves are

$$\begin{aligned}
\frac{\partial \boldsymbol{r}}{\partial r} &= \sin\theta\cos\phi\,\boldsymbol{i} + \sin\theta\sin\phi\,\boldsymbol{j} + \cos\theta\,\widehat{\boldsymbol{z}} \\
\frac{\partial \boldsymbol{r}}{\partial \theta} &= r\cos\theta\cos\phi\,\boldsymbol{i} + r\cos\theta\sin\phi\,\boldsymbol{j} - r\sin\theta\,\widehat{\boldsymbol{z}} \\
\frac{\partial \boldsymbol{r}}{\partial \phi} &= -r\sin\theta\sin\phi\,\boldsymbol{i} + r\sin\theta\cos\phi\,\boldsymbol{j},
\end{aligned}$$

and the scale factors are given by

$$h_r = \left|\frac{\partial \boldsymbol{r}}{\partial r}\right| = 1, \qquad h_\theta = \left|\frac{\partial \boldsymbol{r}}{\partial \theta}\right| = r, \qquad h_\phi = \left|\frac{\partial \boldsymbol{r}}{\partial \phi}\right| = r\sin\theta.$$

The local basis consists of the vectors

$$\begin{aligned}
\widehat{\boldsymbol{r}} &= \sin\theta\cos\phi\,\boldsymbol{i} + \sin\theta\sin\phi\,\boldsymbol{j} + \cos\theta\,\widehat{\boldsymbol{z}} \\
\widehat{\boldsymbol{\theta}} &= \cos\theta\cos\phi\,\boldsymbol{i} + \cos\theta\sin\phi\,\boldsymbol{j} - \sin\theta\,\widehat{\boldsymbol{z}} \\
\widehat{\boldsymbol{\phi}} &= -\sin\phi\,\boldsymbol{i} + \cos\phi\,\boldsymbol{j}.
\end{aligned}$$

See Figure A.3. The local basis is right-handed. The unit vectors do not change with r, but

$$\frac{\partial \widehat{\boldsymbol{r}}}{\partial \theta} = \widehat{\boldsymbol{\theta}}, \qquad \frac{\partial \widehat{\boldsymbol{\theta}}}{\partial \theta} = -\widehat{\boldsymbol{r}}, \qquad \frac{\partial \widehat{\boldsymbol{\phi}}}{\partial \theta} = 0,$$

$$\frac{\partial \widehat{\boldsymbol{r}}}{\partial \phi} = \sin\theta\,\widehat{\boldsymbol{\phi}}, \qquad \frac{\partial \widehat{\boldsymbol{\theta}}}{\partial \phi} = \cos\theta\,\widehat{\boldsymbol{\phi}}, \qquad \frac{\partial \widehat{\boldsymbol{\phi}}}{\partial \phi} = -\sin\theta\,\widehat{\boldsymbol{r}} - \cos\theta\,\widehat{\boldsymbol{\theta}}.$$

The volume element in spherical coordinates is

$$dV = h_r h_\theta h_\phi\,dr\,d\theta\,d\phi = r^2\sin\theta\,dr\,d\theta\,d\phi.$$

The area element on the sphere $r =$ constant is

$$dS_r = h_\theta h_\phi\,d\theta\,d\phi = r^2\sin\theta\,d\theta\,d\phi.$$

The area element on the cone $\theta =$ constant is

$$dS_\theta = h_r h_\phi\,dr\,d\phi = r\sin\theta\,dr\,d\phi.$$

The area element on the half-plane $\phi =$ constant is

$$dS_\phi = h_r h_\theta\,dr\,d\theta = r\,dr\,d\theta.$$

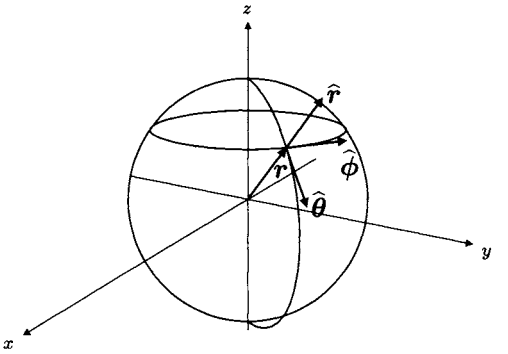

Figure A.3: Spherical coordinate system.

A.4.2 Gradient, divergence and curl

In terms of spherical coordinates, the gradient of the scalar field $f(r, \theta, \phi)$ is expressed by

$$\boldsymbol{\nabla} f(r, \theta, \phi) = \frac{\partial f}{\partial r}\,\widehat{\boldsymbol{r}} + \frac{1}{r}\frac{\partial f}{\partial \theta}\,\widehat{\boldsymbol{\theta}} + \frac{1}{r\sin\theta}\frac{\partial f}{\partial \phi}\,\widehat{\boldsymbol{\phi}}.$$

For spherical coordinates, $h_r = 1$, $h_\theta = r$ and $h_\phi = r\sin\theta$. The divergence of the vector field $\boldsymbol{F} = F_r\,\widehat{\boldsymbol{r}} + F_\theta\,\widehat{\boldsymbol{\theta}} + F_\phi\,\widehat{\boldsymbol{\phi}}$ is

$$\begin{aligned}
\boldsymbol{\nabla}\cdot\boldsymbol{F} &= \frac{1}{r^2\sin\theta}\left[\frac{\partial}{\partial r}(r^2\sin\theta\,F_r) + \frac{\partial}{\partial\theta}(r\sin\theta\,F_\theta) + \frac{\partial}{\partial\phi}(r\,F_\phi)\right] \\
&= \frac{\partial}{\partial r}(r^2\,F_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\,F_\theta) + \frac{1}{r\sin\theta}\frac{\partial F_\phi}{\partial\phi} \\
&= \frac{\partial F_r}{\partial r} + \frac{2}{r}F_r + \frac{1}{r}\frac{\partial F_\theta}{\partial\theta} + \frac{\cot\theta}{r}F_\theta + \frac{1}{r\sin\theta}\frac{\partial F_\phi}{\partial\phi}.
\end{aligned}$$

The curl of $\boldsymbol{F}$ is given by

$$\nabla\times\boldsymbol{F}=\frac{1}{r^2\sin\theta}\begin{vmatrix}\hat{\boldsymbol{r}} & r\hat{\boldsymbol{\theta}} & r\sin\theta\,\hat{\boldsymbol{\phi}}\\ \dfrac{\partial}{\partial r} & \dfrac{\partial}{\partial\theta} & \dfrac{\partial}{\partial\phi}\\ F_r & rF_\theta & r\sin\theta F_\phi\end{vmatrix}$$

$$=\frac{1}{r\sin\theta}\left(\cos\theta\,F_\phi+\sin\theta\,\frac{\partial F_\phi}{\partial\theta}-\frac{\partial F_\theta}{\partial\phi}\right)\hat{\boldsymbol{r}}$$

$$+\frac{1}{r\sin\theta}\left(\frac{\partial F_r}{\partial\phi}-\sin\theta\,F_\phi-r\sin\theta\,\frac{\partial F_\phi}{\partial r}\right)\hat{\boldsymbol{\theta}}$$

$$+\frac{1}{r}\left(F_\theta+r\,\frac{\partial F_\theta}{\partial r}-\frac{\partial F_r}{\partial\theta}\right)\hat{\boldsymbol{\phi}}.$$

A.4.3 Laplace and advection operators

Using the results above, the expressions of the Laplacian operators in spherical coordinates are easily found. The Laplacian of the scalar field $f(r,\theta,\phi)$ assumes the form

$$\nabla^2 f=\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right)+\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\,\frac{\partial f}{\partial\theta}\right)+\frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial\phi^2}$$

$$=\frac{\partial^2 f}{\partial r^2}+\frac{2}{r}\frac{\partial f}{\partial r}+\frac{\cot\theta}{r^2}\frac{\partial f}{\partial\theta}+\frac{1}{r^2}\frac{\partial^2 f}{\partial\theta^2}+\frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial\phi^2},$$

whereas the Laplacian of the vector field $\boldsymbol{F}=F_r\,\hat{\boldsymbol{r}}+F_\theta\,\hat{\boldsymbol{\theta}}+F_\phi\,\hat{\boldsymbol{\phi}}$ is

$$\nabla^2\boldsymbol{F}=\left(\nabla^2F_r-\frac{2F_r}{r^2}-\frac{2}{r^2\sin\theta}\frac{\partial(\sin\theta\,F_\theta)}{\partial\theta}-\frac{2}{r^2\sin\theta}\frac{\partial F_\phi}{\partial\phi}\right)\hat{\boldsymbol{r}}$$

$$+\left(\nabla^2F_\theta-\frac{F_\theta}{r^2\sin^2\theta}-\frac{2\cos\theta}{r^2\sin^2\theta}\frac{\partial F_\phi}{\partial\phi}+\frac{2}{r^2}\frac{\partial F_r}{\partial\theta}\right)\hat{\boldsymbol{\theta}}$$

$$+\left(\nabla^2F_\phi-\frac{F_\phi}{r^2\sin^2\theta}+\frac{2\cos\theta}{r^2\sin^2\theta}\frac{\partial F_\theta}{\partial\phi}+\frac{2}{r^2\sin\theta}\frac{\partial F_r}{\partial\phi}\right)\hat{\boldsymbol{\phi}}.$$

Finally, the advection operators in spherical coordinates assume the form

$$\boldsymbol{a}\cdot\nabla u=a_r\frac{\partial u}{\partial r}+\frac{a_\theta}{r}\frac{\partial u}{\partial\theta}+\frac{a_\phi}{r\sin\theta}\frac{\partial u}{\partial\phi},$$

$$
\begin{aligned}
(\boldsymbol{a}\cdot\boldsymbol{\nabla})\boldsymbol{u} = &\left[(\boldsymbol{a}\cdot\boldsymbol{\nabla})u_r - \frac{a_\theta u_\theta + a_\phi u_\phi}{r}\right]\widehat{\boldsymbol{r}} \\
&+ \left[(\boldsymbol{a}\cdot\boldsymbol{\nabla})u_\theta + \frac{a_\theta u_r - \cot\theta\, a_\phi u_\phi}{r}\right]\widehat{\boldsymbol{\theta}} \\
&+ \left[(\boldsymbol{a}\cdot\boldsymbol{\nabla})u_\phi + \frac{a_\phi u_r + \cot\theta\, a_\phi u_\theta}{r}\right]\widehat{\boldsymbol{\phi}}.
\end{aligned}
$$